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Jeffrey Cheah Foundation
Nurturing the Seeds of Wisdom

Topic 05:
Scheduler in Cluster Computing

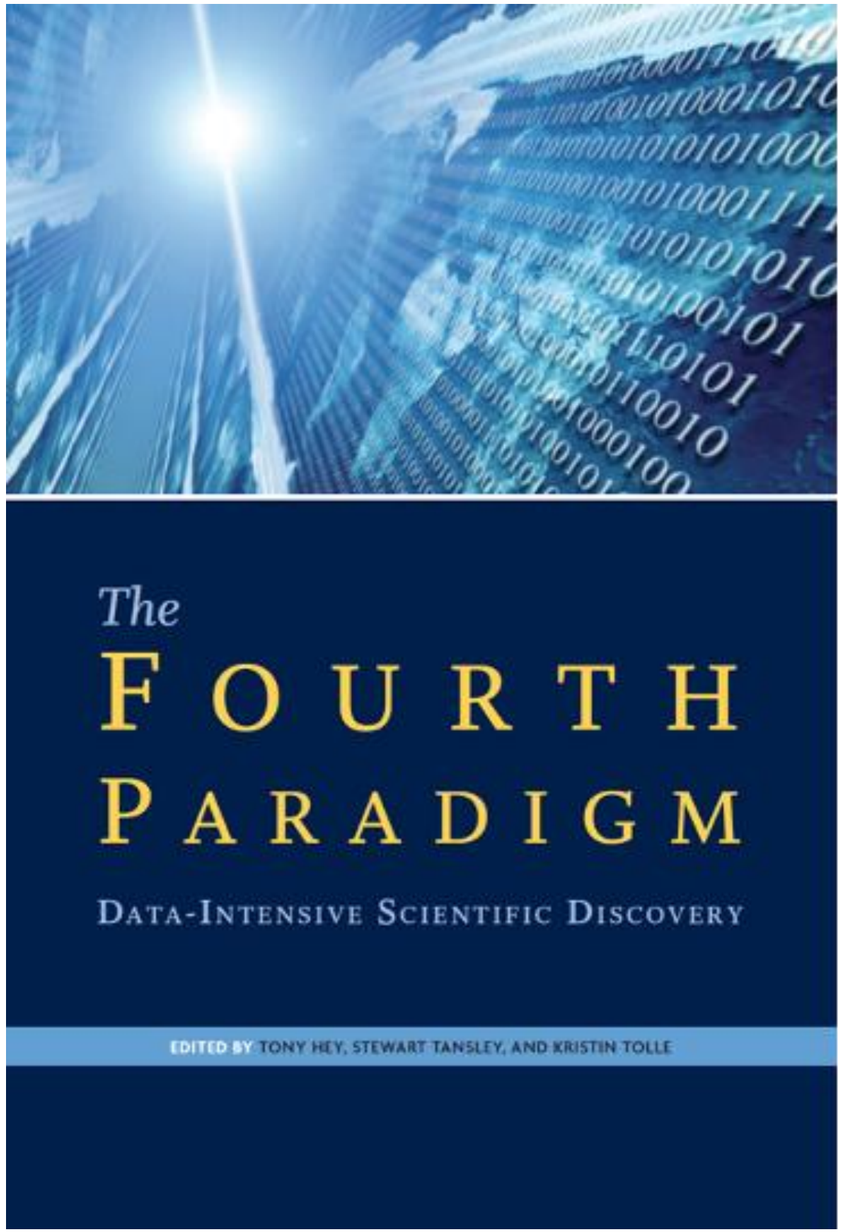
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NET3204
Distributed Systems

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Learning Outcome

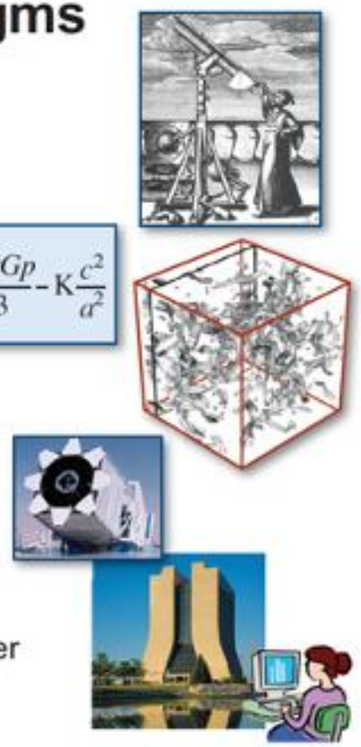
- At the end of this lecture, you will be able to:
  - Explain the concept of cluster computing
  - Describe one of the popular cluster computing platform, i.e. HTCondor



Science Paradigms

- Thousand years ago:
  science was **empirical**
  describing natural phenomena
- Last few hundred years:
  theoretical branch
  using models, generalizations
- Last few decades:
  a **computational** branch
  simulating complex phenomena
- Today: **data exploration** (eScience)
  unify theory, experiment, and simulation
  - Data captured by instruments or generated by simulator
  - Processed by software
  - Information/knowledge stored in computer
  - Scientist analyzes database/files using data management and statistics

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{4\pi G\rho}{3} - \mathrm{K}\frac{c^2}{a^2}$$





Govt aims for Malaysia to be top regional AI hub

By Qistina Sallehuddin - November 6, 2023 @ 2:48pm

NEW STRAITS TIMES



MALAYSIA

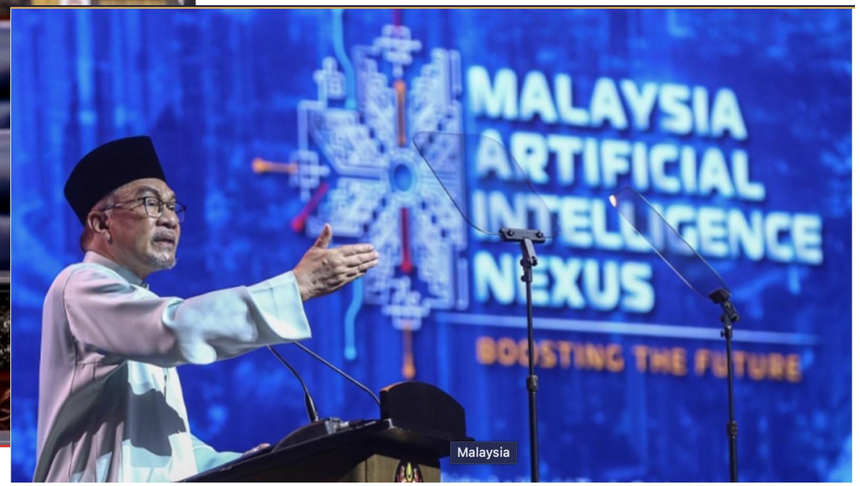
PM Anwar launches national AI office to turn Malaysia into regional digital hub

PM Anwar launches national AI office to turn Malaysia into regional digital hub



Moving forward with AI: How Malaysia can grow into an AI-savvy nation with the right talent and innovation

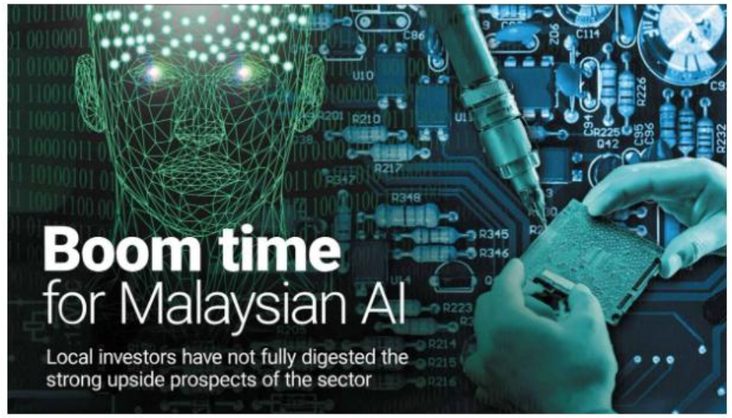
By ANGELIN YEOH and CHRISTOPHER FAM



malaymail.com

ASIA 2 min. Read

Malaysia Launches MY AI NEXUS Programme





Chang (middle) receiving a token of appreciation from Low (left) and ACCCIM AI Unplugged organising chairman David Lim at the conference in Kuala Lumpur last week. — FAIHAN GHANI/The Star

As the world gears up for artificial intelligence to change the way we work, communicate and learn, Malaysia is working towards nurturing more AI talent.

his Week in Asia / Economics

Malaysia pushes AI upskilling courses to 'future-proof' economy: 'data is the new oil'

Malaysia could generate as much as US$113.4 billion – a quarter of its GDP – through the embrace of generative AI

Renewable energy and AI are key themes of PM Anwar Ibrahim's administration to wean Malaysia off its reliance on its depleting oil reserves

Listen to this article

Hadi Azmi + FOLLOW

Published: 5:00pm, 16 Jan, 2024

South China Morning Post

Malaysia towards a top AI Nation

Malaysia's aspiration of being among the top 20 countries in terms of AI

Embracing AI technology is crucial for societal advancement



Talent

The people to drive the innovation and make the change



Technology

The infrastructure including hardware, software and services as the enabler for AI adoption



Data

AI needs good data, i.e. quality and quantity

"We cannot, for example, seek to make Malaysia a success as a digital nation while being bound by old and outdated mindsets,"

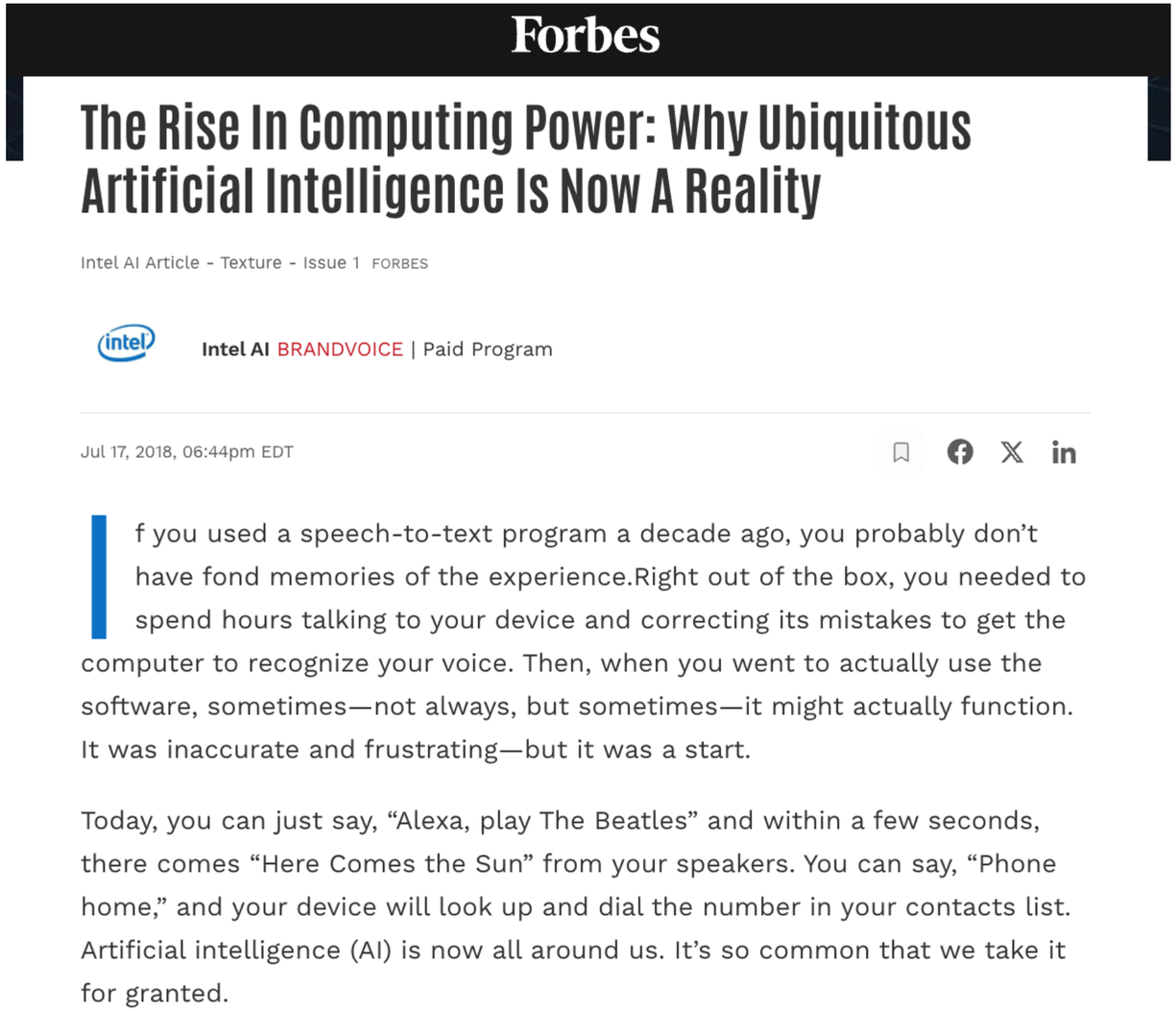
~ PM Datuk Seri Anwar Ibrahim

Exa-scale computers: a Quintillion Calculations /s

New “exa-scale” supercomputers will bring breakthroughs in science.

Why are algorithms that were basically invented in the 1950s through the 1980s only now causing such a transformation of business and society?

The answer is simple: Technology has finally become powerful enough to enable the promise of AI.



Forbes

The Rise In Computing Power: Why Ubiquitous Artificial Intelligence Is Now A Reality

Intel AI Article - Texture - Issue 1 FORBES

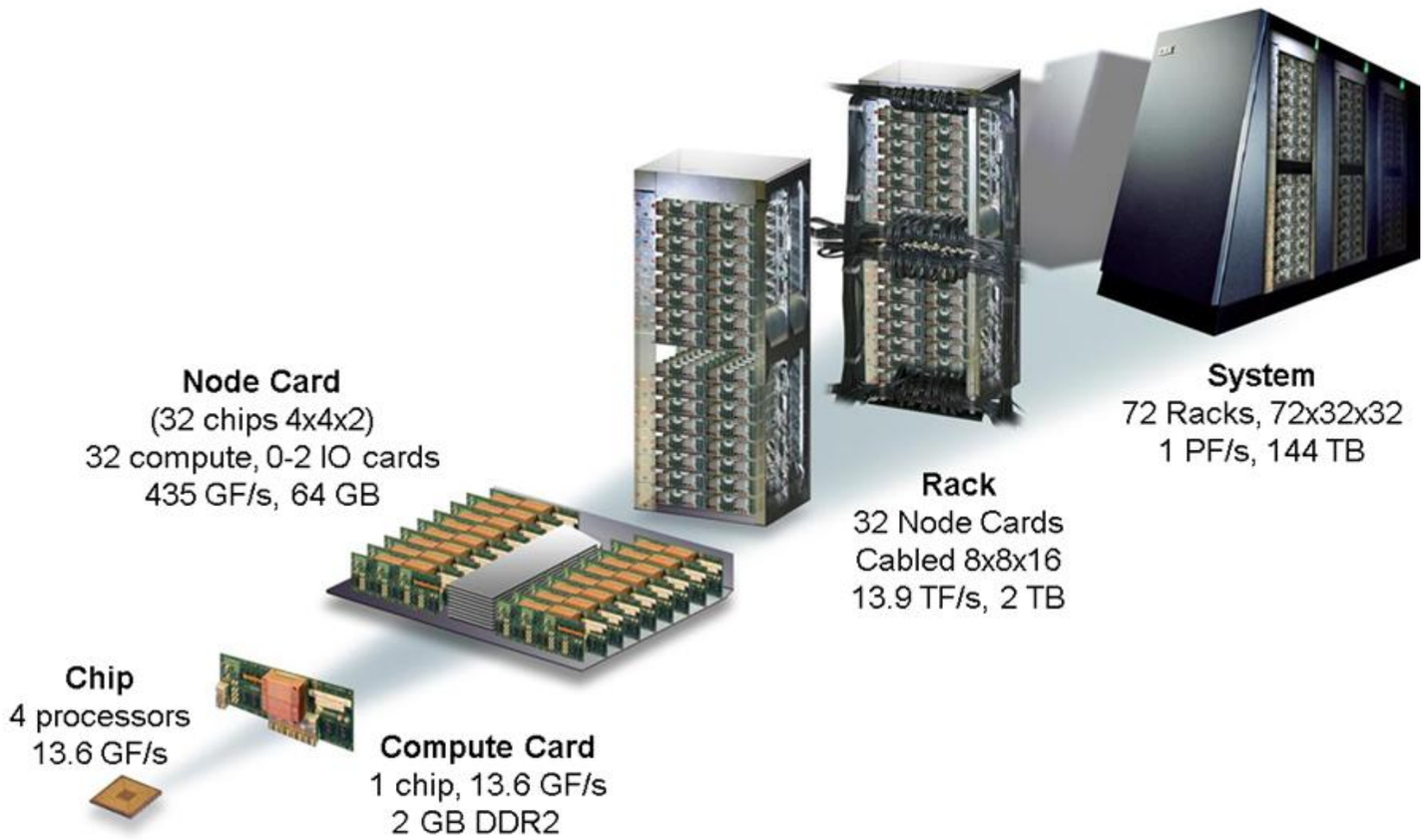
Intel AI BRANDVOICE | Paid Program

Jul 17, 2018, 06:44pm EDT

If you used a speech-to-text program a decade ago, you probably don’t have fond memories of the experience.Right out of the box, you needed to spend hours talking to your device and correcting its mistakes to get the computer to recognize your voice. Then, when you went to actually use the software, sometimes—not always, but sometimes—it might actually function. It was inaccurate and frustrating—but it was a start.

Today, you can just say, “Alexa, play The Beatles” and within a few seconds, there comes “Here Comes the Sun” from your speakers. You can say, “Phone home,” and your device will look up and dial the number in your contacts list. Artificial intelligence (AI) is now all around us. It’s so common that we take it for granted.

From Chip to Supercomputer



Source: IBM System Blue Gene Solution: Blue Gene/P Application Development (https://www.redbooks.ibm.com/redbooks/pdfs/sg247287.pdf)

What is Cluster Computing?

- A group of interconnected computers (nodes) working together as a single powerful system, creating a unified computing resource far greater than the sum of its parts.
- By pooling resources, clusters enable high-performance computing capabilities that transform how we approach complex problems.
- This architecture provides the foundation for tackling intricate scientific challenges that single machines simply cannot handle efficiently or economically.

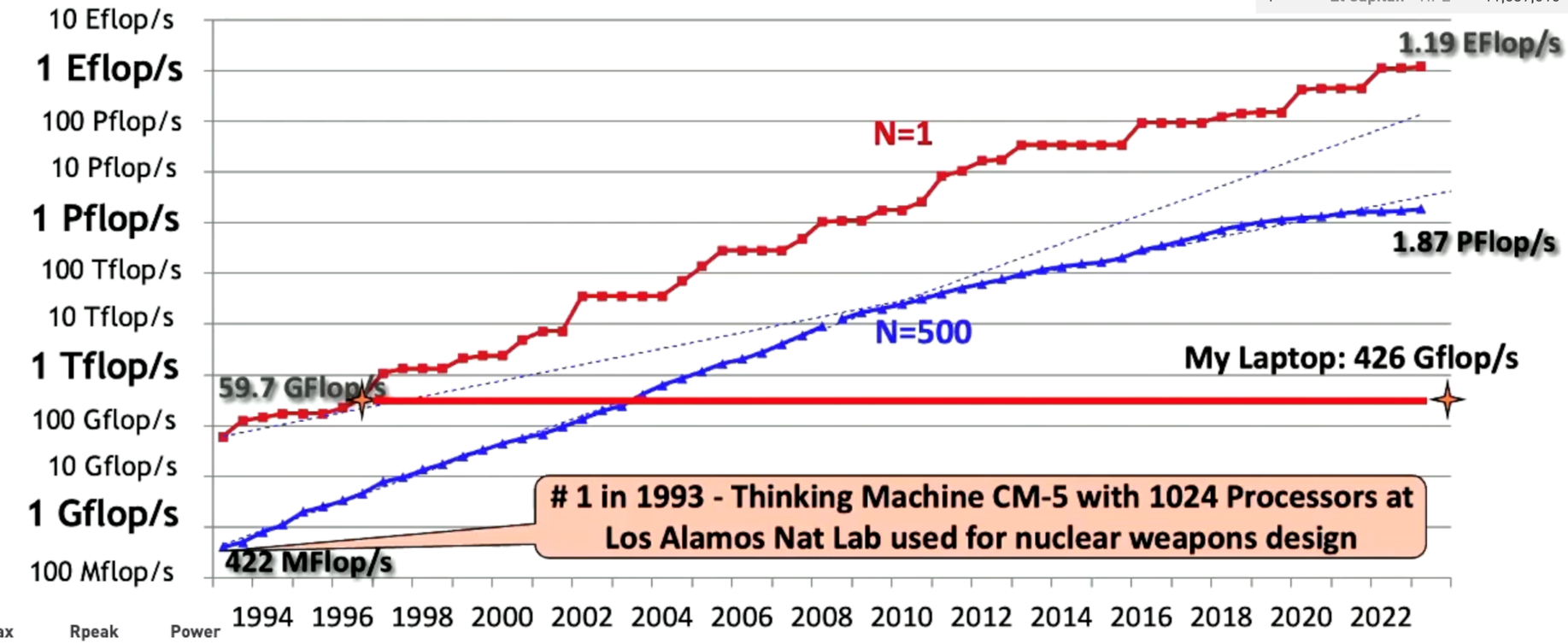
"Now, what used to be thought of as supercomputers are inside smartphones. They cost a million times less, are a million times faster and have a million times as much memory."

-Richard Mark Soley, Ph.D., executive director, Industrial Internet Consortium

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Performance Development of HPC over the Last 30 Years from the Top500



#1 in 2025 June

Rank	System	Cores	Rmax (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
1	El Capitan - HPE	11,039,616	1,742.00	2,746.38	29,581

#1 in 1993 June

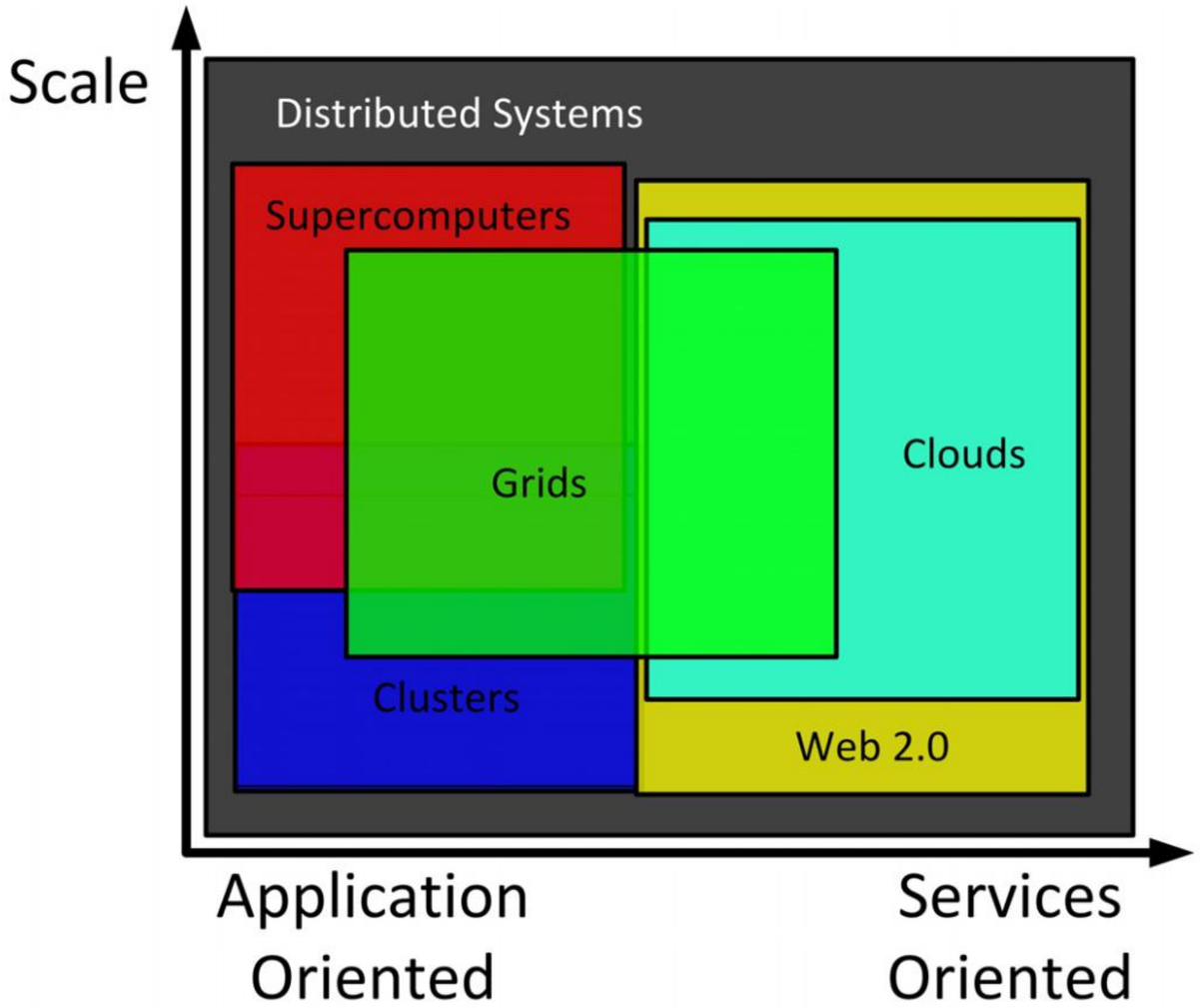
Rank	System	Cores	Rmax (GFlop/s)	Rpeak (GFlop/s)	Power (kW)
1	CM-5/1024,	1,024	59.70	131.00	

https://www.top500.org/

Cluster Computing in Scientific Discovery

- Climate Modelling
  - Accelerates complex simulations predicting weather patterns and climate change scenarios decades into the future
- Genomics Research
  - Processes vast genetic datasets to unlock the secrets of disease, evolution, and personalised medicine
- Astrophysics
  - Analyses petabytes of cosmic data from observatories, revealing the universe's deepest mysteries

Distributed Computing



Source: I. Foster, Y. Zhao, I. Raicu, and S. Lu. Cloud computing and grid computing 360-degree compared.

Cluster Architecture Overview

- Compute Nodes
  - Individual computers equipped with CPUs/GPUs, memory, and local storage, forming the processing backbone
- High-Speed Network
  - InfiniBand or Ethernet interconnects enable rapid communication between nodes with minimal latency
- Resource Manager

  Topic 05
  - Orchestrates job distribution, monitors system health, and optimises resource allocation across the cluster
- Shared Storage

  Topic 06
  - Distributed file systems provide unified access to data, ensuring all nodes can retrieve and store information efficiently

The Role of Cluster Schedulers

- Schedulers are the intelligent orchestrators of cluster computing, managing job queues, allocating resources, and strategically assigning tasks to nodes based on availability and priority.
- Their primary objectives include maximising cluster utilisation, minimising user wait times, and ensuring fair access across multiple research teams and projects.
- Modern schedulers elegantly handle diverse workloads—from quick interactive tasks and long-running batch jobs to always-on services—whilst adapting to changing demands in real time.

The Condor Project

- established in 1985
- research and development of distributed high throughput computing



2012



Miron Livny



1988



2025

Miron's quotes

- the problems we are facing are fundamental and silver bullet technologies are pipe-dreams, for computing in general and distributed computing in particular
- Bringing the right data to the processing unit and shipping the results back to the consumer is a fundamental (hard) problem.

Condor is

- High throughput computing, instead of high performance computing
  - less concerned about instantaneous computing power, but the amount of computing that can be harnessed over a month or a year
- all about match making
  - bring together the one who requests for computing power with those who offer them
  - "*Condor - a hunter of idle workstations*." [1988]

Definitions

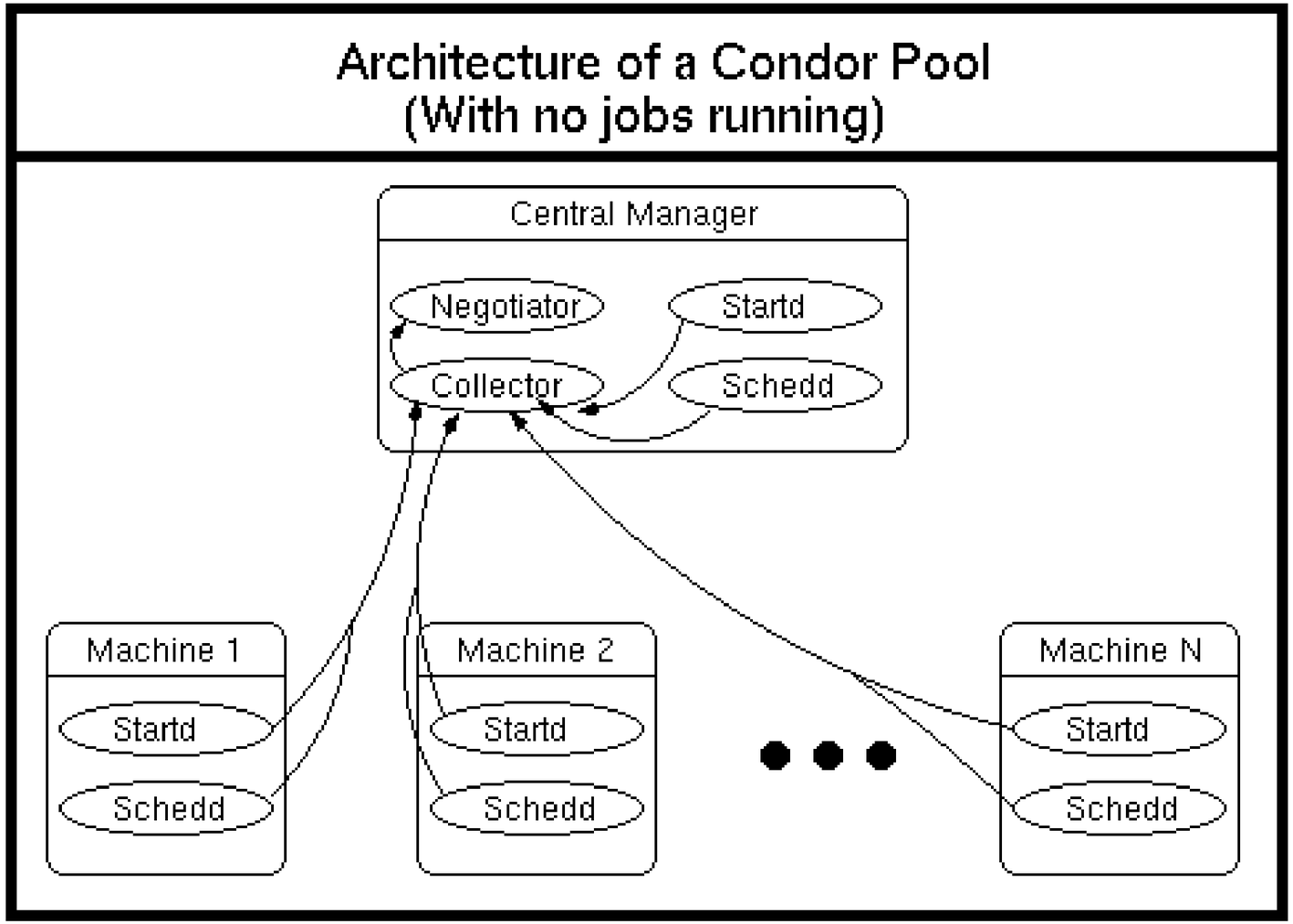
- Job
  - the Condor representation of a piece of work
  - Condor's quanta of work
  - Like a Unix process, and can be an element of a workflow
- ClassAd
  - Condor's internal data representation
- Machine or Resource
  - computers that can do the processing
- Pool
  - A collection of computers used by Condor

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More definitions

- Match Making
  - Associating a job with a machine resource
- Central Manager
  - Central repository for the whole pool
  - Does match making
- Submit Host
  - The computer from which jobs are submitted to Condor
- Execute Host
  - The computer that runs a job

Condor daemons



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Condor daemons

- condor_master:
  - This program runs constantly and ensures that all other parts of Condor are running. If they hang or crash, it restarts them.
- condor_collector:
  - This program is part of the Condor central manager. It collects information about all computers in the pool as well as which users want to run jobs. It is what normally responds to the condor_status command.
- condor_negotiator:
  - This program is part of the Condor central manager. It decides what jobs should be run where.
- condor_startd:
  - If this program is running, it allows jobs to be started up on this computer--that is, your computer is an "execute machine". This advertises your computer to the central manager (more on that later, but in this case it's also your computer) so that it knows about this computer. It will start up the jobs that run.
- condor_schedd
  - If this program is running, it allows jobs to be submitted from this computer—that is, your computer is a "submit machine". This will advertise jobs to the central manager so that it knows about them. It will contact a condor_startd on other execute machines for each job that needs to be started.
- condor_shadow (Not shown above)
  - For each job that has been submitted from this computer, there is one condor_shadow running. It will watch over the job as it runs remotely. In some cases it will provide some assistance. You may or may not see any condor_shadow processes running, depending on what is happening on the computer when you try it out.

Setting up your first Condor Pool

- [follow the QuickStart Guide at https://htcondor.readthedocs.io/en/latest/getting-htcondor/admin-quick-start.html to set up the 3 Nodes cluster]
  - A typical set up is:
    - a Condor Central Manager, a.k.a. Condor Host, to manage the entire Condor Pool, e.g. collecting ClassAd from all executors, received job request from submission host, and perform matchmaking.
    - a Submission host, for managing the job queue, and
    - a couple of Execution Hosts, the worker nodes.

A simplified installation step

- first, as the adminisitrator of your condor pool, follow the instruction at https://htcondor.readthedocs.io/en/latest/getting-htcondor/admin-quick-start.html
- prepare 3 machines from your AWS
- edit the /etc/hosts with the hosts information
- at the condor_host machine , run this: curl -fsSL https://get.htcondor.org | sudo GET_HTCONDOR_PASSWORD="password" /bin/bash -s -- --no-dry-run --central-manager Condor_Host
- at the execution_host machine , run this: curl -fsSL https://get.htcondor.org | sudo GET_HTCONDOR_PASSWORD="password" /bin/bash -s -- --no-dry-run --execute Condor_Host
- at the submission_host machine , run this: curl -fsSL https://get.htcondor.org | sudo GET_HTCONDOR_PASSWORD="password" /bin/bash -s -- --no-dry-run --submit Condor_Host
- restart the machine with this command: sudo systemctl restart condor
- check the status of the daemons using this command: sudo systemctl status condor
- You can now see your condor pool with this command: condor_status

Sample output from executing condor_q and condor_status

```
ubuntu@ip-10-135-46-41:~/condor$ condor_q

-- Submitter: ip-10-135-46-41.ap-southeast-1.compute.internal : <10.135.46.41:42992> : ip-10-135-
46-41.ap-southeast-1.compute.internal
 ID      OWNER            SUBMITTED     RUN_TIME ST PRI SIZE CMD
   2.0   ubuntu          4/17 16:59   0+00:00:00 I  0   0.0  simple 4 10
   2.1   ubuntu          4/17 16:59   0+00:00:00 I  0   0.0  simple 4 11
   2.2   ubuntu          4/17 16:59   0+00:00:00 I  0   0.0  simple 4 12

3 jobs; 0 completed, 0 removed, 3 idle, 0 running, 0 held, 0 suspended

ubuntu@ip-10-135-46-41:~/condor$ condor_status

Name               OpSys      Arch   State     Activity LoadAv Mem   ActvtyTime

slot1@ip-10-135-46 LINUX      X86_64 Claimed   Busy     0.020  3725  0+00:00:04
slot2@ip-10-135-46 LINUX      X86_64 Claimed   Busy     0.000  3725  0+00:00:05
                     Total Owner Claimed Unclaimed Matched Preempting Backfill

        X86_64/LINUX     2     0       2         0       0          0        0

               Total     2     0       2         0       0          0        0
```


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Let's submit our first Condor Job

- [follow the User's QuickStart Guide at https://htcondor.readthedocs.io/en/latest/users-manual/quick-start-guide.html]
  - Once your Condor pool is up and running, you can try to create a job and submit to your condor pool.
  - You need:
    - A job - the executable/[program that you plan to execute on condor. For instance, a bash script.
    - A job description file - a text file specifying the detail of the job, including the input and output files.

Sample job

- [taken from the User's QuickStart Guide]

```
#!/bin/bash
# file name: sleep.sh

TIMETOWAIT="6"
echo "sleeping for $TIMETOWAIT seconds"
/bin/sleep $TIMETOWAIT
```

Job description file

- [taken from the User's QuickStart Guide]

```
# sleep.sub -- simple sleep job
executable = sleep.sh
log        = sleep.log
output     = sleep.out
error    = sleep.err

should_transfer_files     = Yes
when_to_transfer_output   = ON_EXIT
request_cpus     = 1
request_memory     = 512M
request_disk     = 1G

queue
```


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Job description file (continue)

- The first few lines list the input and output files of the job.
- The second part specific the handling of the files, either staging in, or by using shared file system
- The third part shows the specification of the requested resources to run the job. In the example, it requires a single CPU with 512MB RAM.
- The last part of the script is a keyword **queue**, which instruct the Submission Host to submit this script to the Condor Manager.
- Then, you just need to submit this using the command:

```
condor_submit sleep.submit
```

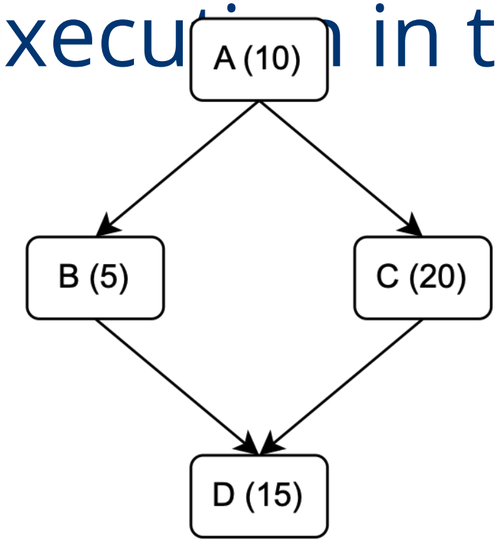

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Check the log file after execution

```
ubuntu@ip-10-135-46-41:~/condor$ more sleep.log
000 (001.000.000) 04/17 16:56:30 Job submitted from host: <10.135.46.41:42992>
...
001 (001.000.000) 04/17 16:56:31 Job executing on host: <10.135.46.41:48957>
...
006 (001.000.000) 04/17 16:56:35 Image size of job updated: 10
  0  -  MemoryUsage of job (MB)
  0  -  ResidentSetSize of job (KB)
...
005 (001.000.000) 04/17 16:56:35 Job terminated.
  (1) Normal termination (return value 0)
    Usr 0 00:00:00, Sys 0 00:00:00  -  Run Remote Usage
    Usr 0 00:00:00, Sys 0 00:00:00  -  Run Local Usage
    Usr 0 00:00:00, Sys 0 00:00:00  -  Total Remote Usage
    Usr 0 00:00:00, Sys 0 00:00:00  -  Total Local Usage
  56  -  Run Bytes Sent By Job
  8531  -  Run Bytes Received By Job
  56  -  Total Bytes Sent By Job
  8531  -  Total Bytes Received By Job
  Partitionable Resources :    Usage  Request
     Cpus                 :                1
     Disk (KB)            :       17       17
     Memory (MB)          :        0        0
...
```

Condor DAGMan

- The workflow is represented as a Directed Acyclic Graph (DAG).
- HTCondor has a component called DAGMan that handles the DAG execution in the platform.



- A DAGMan workflow is described in a DAG input file.
- The input file specifies the nodes of the DAG as well as the dependencies that order the DAG, as shown below:

```
JOB A A.sub
JOB B B.sub
JOB C C.sub
JOB D D.sub
PARENT A CHILD B C
PARENT B C CHILD D
```

- The PARENT A CHILD B C indicate that task A needs to be executed before task B and task C.

Submitting your DAG

- [follow the DAGMan guide at https://htcondor.readthedocs.io/en/latest/automated-workflows/dagman-introduction.html#describing-workflows-with-dagman]
- You can submit the DAG file using this command:

```
condor_submit_dag diamond.dag
```


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Summary

- What did I learn in this module?
  - The concept of cluster computing
  - The HTCondor platform
  - Creating a HTCondor Pool
  - Submitting a HTCondor job, and a workflow DAG

Acknowledgement

The figures and contents used in this lecture are adopted and adapted from:

- van Steen, M. and Tanenbaum A.S. (2023). Distributed Systems, 4th Edition. distributed-systems.net.
- Colouries, G., Dollimore, J., Kindberg, T and Blair, G. (2021). Distributed Systems: Concepts and Design, 5th Edition. Pearson.
- Web references.

THANK YOU

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